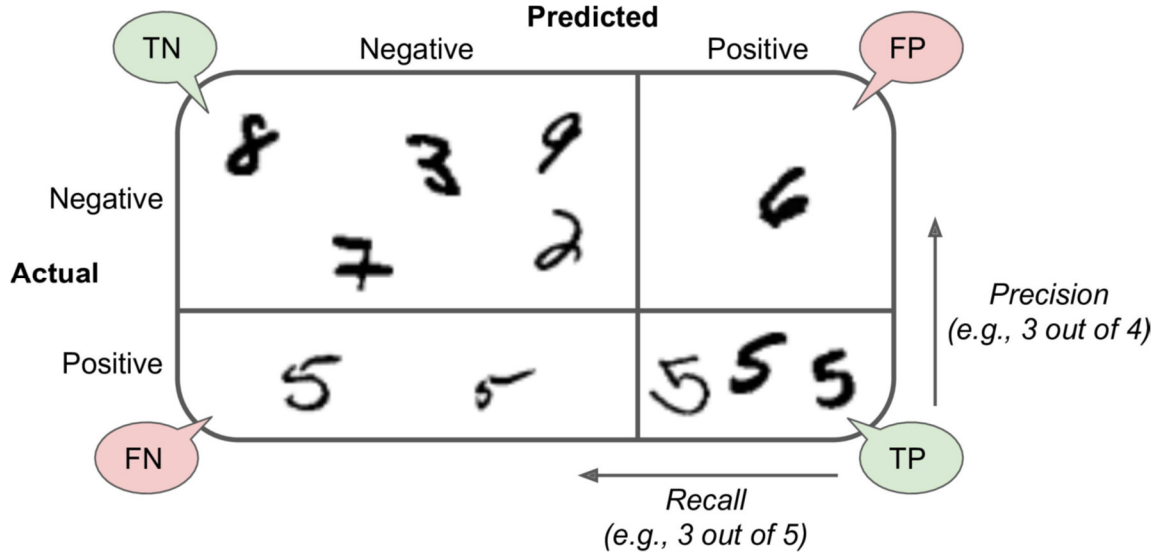


Performance Measures for Evaluating a Binary Classifier

- **Accuracy:** Ratio of correct predictions on all input data
- **Confusion Matrices**



- **Precision:** Accuracy of the positive prediction

$$\text{precision} = \frac{TP}{TP + FP}$$

- **Recall** (called sensitivity or the true positive rate (**TPR**)): Ratio of positive instance that are correctly detected by the classifier

$$\text{recall} = \frac{TP}{TP + FN}$$

- **F1 Score:** Harmonic mean of precision and recall

$$F_1 = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2TP}{TP + FN + FP}$$

- **Receiver Operating Characteristic (ROC) curve**

- Plot **TPR** against **FPR**

$$FPR = \frac{FP}{FP + TN} = 1 - TNR = 1 - \text{specificity}$$

- **AUC:** Area Under the Curve

Exercise: Find the performance measures for the following confusion matrices

$$\begin{bmatrix} 90 & 10 \\ 50 & 150 \end{bmatrix}$$

- Accuracy = $(90 + 150) / (100 + 200) = 0.8$
- Precision = $150 / (150 + 10) = 0.9375$
- Recall = $150 / (150 + 50) = 0.75$
- F1 score = $2 / (1 / 0.9375 + 1 / 0.75) = 0.83$
- TPR = $150 / (150 + 50) = 0.75$
- FNR = $1 - \text{TPR} = 0.25$
- TNR = $90 / (90 + 10) = 0.9$
- FPR = $1 - \text{TNR} = 0.1$

Note: in the code, use the “sklearn” library to evaluate the performance, such as

```
from sklearn.metrics import confusion_matrix, precision_score, recall_score, f1_score,  
precision_recall_curve, roc_curve, roc_auc_score
```